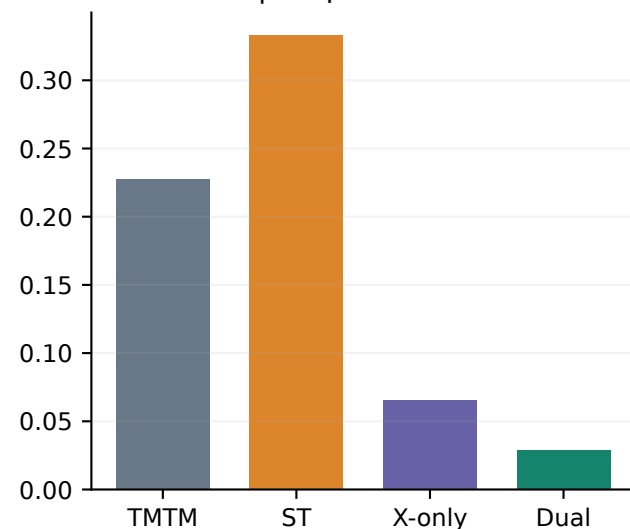
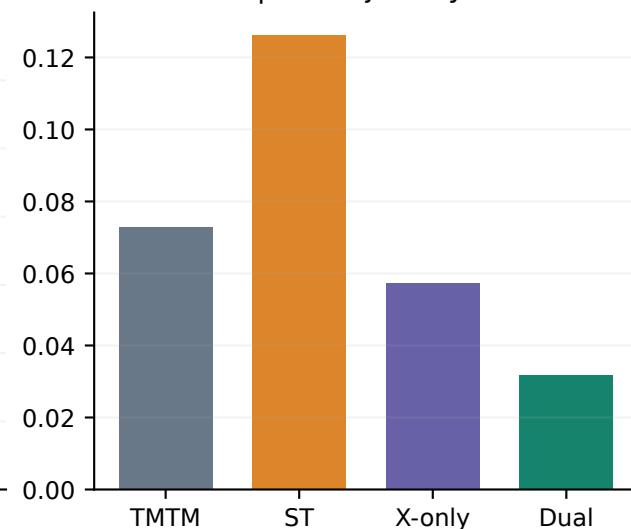


Public-data validation: reference accuracy, geometric trade-off, and computational cost
Single-view 2D readout receives true birth-Y held fixed; timing includes layout, rendering and validation

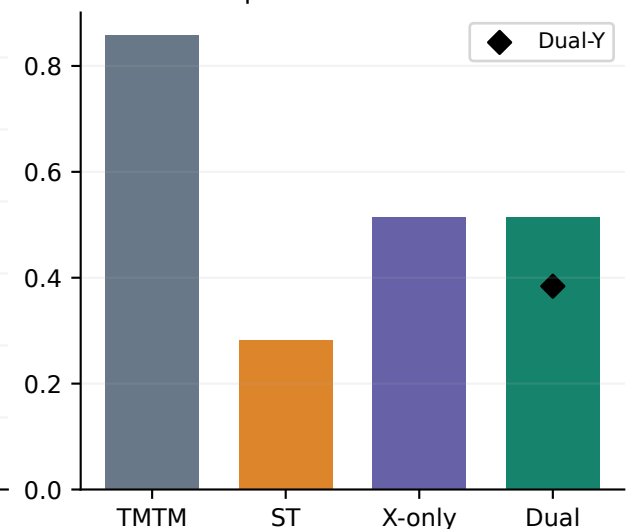
RING | 2D position NRMSE



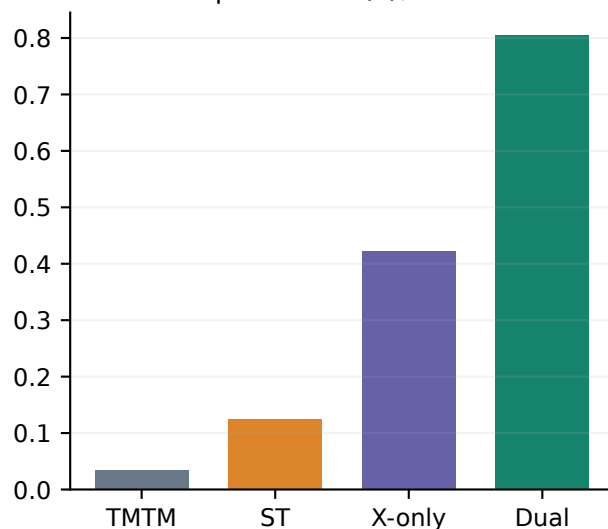
RING | 2D trajectory NMAE



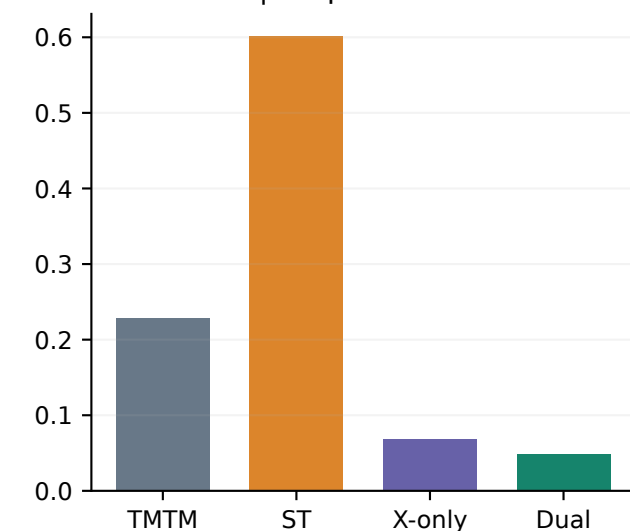
RING | 1D distance NRMSE



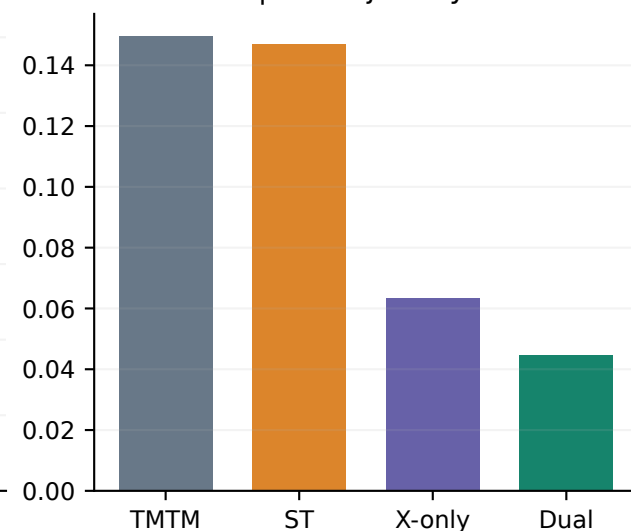
RING | Runtime (s), median 3



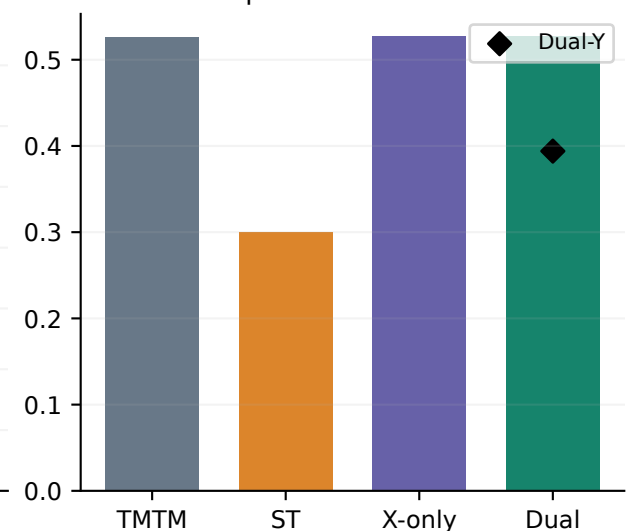
ERA5 | 2D position NRMSE



ERA5 | 2D trajectory NMAE



ERA5 | 1D distance NRMSE



ERA5 | Runtime (s), median 3

